Structure-Aware Phylogenomics of the *Plasmodium vivax* VIR Multigene Family

Generated from Loom notebook, May 2026

2026-05-28

Abstract

We apply the FoldMason structural phylogenomics pipeline (Gilchrist et al., 2026, *Science*) to the *Plasmodium vivax* VIR multigene family across three modern long-read reference assemblies (PvP01, PvW1, PvPAM). Structural clustering with Foldseek + ProstT5 collapses the 2,592 VIR proteins into 73 structural classes – roughly half as many as sequence-based MCL clusters – with a single class (FS_001) accounting for 46% of the family. A maximum- likelihood phylogeny of 338 representative sub-sequences from FS_001 shows no detectable geographic structure: parsimony scores of "genome of origin" mapped onto each tree are indistinguishable from random label shuffles. The data are consistent with ancient VIR diversification predating the geographic separation of *P. vivax* lineages, plus ongoing recombination and gene conversion that keeps the family panmictic across continents.

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0.1 1. The FoldMason paper

Gilchrist, Mirdita, and Steinegger (2026, *Science*) introduced FoldMason, a progressive multiple structural alignment (MSTA) method that operates simultaneously on amino acid (AA) sequences and on the 3Di structural alphabet developed for Foldseek. The 3Di alphabet encodes the local backbone geometry of every residue as one of twenty letters, so two proteins that share a fold but no detectable sequence identity will still align in 3Di. FoldMason builds a guide tree from all-versus-all 3Di+AA pairwise alignments and then performs progressive alignment from the leaves to the root, producing two output alignments in lockstep: one over the AA alphabet and one over the 3Di alphabet, with identical column coordinates.

A central demonstration in the paper is that FoldMason MSTAs are good enough to recover known phylogenetic relationships even between proteins that are no longer alignable by sequence-based methods. The authors took the Flaviviridae glycoprotein dataset assembled by Mifsud et al. (2024), which spans the E, E1, and E2 envelope glycoprotein families (roughly thirty to fifty proteins per family, with pairwise sequence identity in some cases well below the so-called twilight zone of 20-25%). For each family they ran FoldMason `easy-msa` to produce AA and 3Di alignments, trimmed columns containing 35% or more gaps with trimAl, ran ModelFinder to select an AA

substitution model and used the published Garg & Hochberg 2025 3Di substitution matrix for the 3Di alignment, and inferred three maximum-likelihood trees per family with IQ-TREE 2 and 1000 ultrafast bootstrap replicates: one tree from AA alone, one from 3Di alone, and one from a partitioned analysis of concatenated AA+3Di. Tree similarity to the Mifsud et al. reference trees was measured with quartet similarity and Robinson-Foulds distance from the Quartet R package. Across the three glycoprotein families, FoldMason-derived trees were highly concordant with the manually constructed reference phylogenies, with much of the gain attributable to the 3Di partition recovering relationships invisible to sequence alone.

0.2 2. What we set out to do

We wanted to apply this same structure-aware phylogenomic pipeline to the largest and most poorly understood gene family in the *Plasmodium vivax* genome: the VIR (or *vir*) family of variant surface antigens, part of the broader PIR (*Plasmodium* interspersed repeat) superfamily. Three considerations motivated the work. First, VIRs are extremely sequence-divergent, so sequence-only methods cluster them into many small fragments that almost certainly do not reflect the true family structure. Second, Harrison and Higgins (2020) had shown structurally that PIRs share a single conserved fold despite that sequence divergence, which is exactly the situation in which structural alignment should outperform sequence alignment. Third, three high-quality long-read reference genomes had recently become available from geographically distant *P. vivax* isolates – PvP01 from Papua, Indonesia; PvW1 from Thailand; PvPAM from the Peruvian Amazon – making it possible to ask not only how the VIR family is organized but whether its organization is shared across continents or evolved independently after the geographic separation of *P. vivax* lineages.

Our plan, designed to mirror the FoldMason paper as closely as the data allowed, had three stages. Stage one (Plan A) was a sequence-based clustering with MMseqs2 and Markov Clustering to provide a baseline reference point. Stage two (Plan B) was structure-aware clustering of the same set of proteins with Foldseek using the ProstT5 protein language model to predict 3Di alphabet sequences directly from amino acid sequence (so we did not need to compute any actual three-dimensional structures). Stage three (Plan C) was phylogenetic analysis with FoldMason and IQ-TREE following the procedure used in the FoldMason paper for the glycoproteins, applied twice: once at the family level on the structural cluster representatives, and once within the dominant structural cluster to look for geographic signal among individual VIR sequences.

0.3 3. What is known about PIR / VIR genes

The PIR superfamily is the largest multigene family in essentially every *Plasmodium* species sequenced to date, including the rodent malarias (where it has the lineage-specific names *cir*, *bir*, *yir*), the simian parasites *P. knowlesi* (*kir*) and *P. cynomolgi* (*cyir*), the human-infecting *P. vivax* (*vir*), and *P. ovale* and *P. malariae*. The dramatic exception is *P. falciparum*, which has no PIR genes at all; instead, its functionally analogous variant antigen family is *var*. PIR genes are concentrated in the subtelomeric regions of every chromosome, where the high local recombination rate appears to act as an evolutionary engine for rapid duplication, gene conversion, and pseudogenization. Their typical gene architecture, conserved across species, is a three-exon structure: a short first exon encoding a signal peptide that directs the protein into the secretory pathway, a much longer second exon that encodes a highly variable extracellular domain responsible for antigenic variation, and a short third exon encoding a transmembrane anchor.

The number of copies per genome varies enormously between species and is exquisitely sensitive

to assembly quality. The original short-read Salvador-I reference of *P. vivax* (Carlton et al., 2008) reported 346 *vir* genes organized into twelve subfamilies (A through L); long-read assemblies have since revealed that the true *vir* repertoire of *P. vivax* contains well over a thousand copies, with much of the early missing complement hidden in unassembled subtelomeric contigs. The PvP01 reference (Auburn et al., 2016) reported approximately 1,200 *vir* genes; the PvW1 reference (Minassian et al., 2021) reports 1,145; the PvPAM reference (Vaughan et al., 2023) reports 975 functional copies plus 99 pseudogenes. Notably, more than 600 of the PvPAM *vir* genes share no direct one-to-one orthology with PvP01, an observation that has been used to argue that the family is undergoing rapid, lineage-specific expansion.

Despite their abundance, the functions of individual PIR/VIR proteins remain poorly characterized. Several broad roles are reasonably well-supported in the literature. They are exported to the surface of the infected red blood cell, where they contribute to antigenic variation: switching the surface PIR repertoire allows the parasite to escape the host's antibody response and establish chronic infections. They appear to mediate cytoadherence, binding to receptors on the endothelium of small blood vessels and thereby allowing the infected red cell to avoid clearance by the spleen. They also play a role in rosetting, the clumping of infected and uninfected red cells together, which is associated with severe malaria. In *P. vivax*, transcriptomic studies have associated changes in *vir* expression with severe clinical outcomes including hepatic dysfunction, renal failure, and cerebral malaria, even though *P. vivax* was historically considered a "benign" form of malaria. Several groups have also reported that specific *vir* subfamilies are expressed at different parasite life-cycle stages, suggesting that the variation within the family is functionally partitioned rather than purely redundant.

The structural biology of the family was substantially clarified by Harrison and Higgins (2020), who solved the first crystal structure of a PIR ectodomain, using a member of the family from the rodent parasite *P. chabaudi chabaudi*. The structure (PDB: 6ZYV) showed a small, compact, mostly alpha-helical fold that the authors argued was conserved across the entire PIR superfamily, despite the extreme sequence variation. This is precisely the situation that motivates structural rather than sequence-based phylogenomics: the proteins share a fold, but for many pairs the sequence similarity is below the level at which sequence alignment is reliable.

0.4 4. Analysis

0.4.1 4.1. Extracting the VIR proteins

For each of the three reference genomes (PvP01, PvW1, PvPAM) we downloaded the full annotated proteome FASTA from PlasmoDB release 71. We attempted to extract VIR proteins by parsing the gene-product annotations in the FASTA headers, but found that PvW1 in particular only labels about fifty of its sequences as "VIR protein" – most of its expected ~1,100 *vir* genes are labelled "unspecified product" in the PlasmoDB release. We therefore used the canonical Pfam profile HMM for the *vir* domain (`PF05795`, "Plasmodium_Vir") with `hmmer3` to identify VIR proteins by sequence-profile similarity rather than annotation text. This recovered 931 VIR proteins from PvP01, 748 from PvW1, and 985 from PvPAM, for a combined dataset of 2,664 proteins. After accounting for sequences that later dropped out during clustering (those that didn't form pairs at our coverage threshold), 2,592 proteins entered the downstream analyses.

0.4.2 4.2. Sequence-only baseline (Plan A)

We first asked what structure the family takes when one uses only the amino acid sequences. We ran an all-versus-all sequence comparison of the 2,664 VIR proteins with MMseqs2 at default settings (E-value 1e-5, sequence identity threshold 30%, mutual coverage 50%), converted the resulting bitscore matrix into the ABC format expected by the Markov Clustering Algorithm (MCL), and ran MCL with an inflation parameter of 2.0. This produced 146 sequence-based subfamilies containing three or more members. The largest contained 191 proteins, the next 182, and the size distribution was relatively flat: no single cluster dominated. When we cross-tabulated the cluster membership against genome of origin, 132 of the 146 subfamilies (90%) contained members from all three genomes, indicating that the family's sequence-level structure is largely conserved across continents at the resolution of the clustering. We will return to this number after the structural clustering, because the sequence-only picture is misleading.

0.4.3 4.3. Structure-aware clustering with Foldseek + ProstT5 (Plan B)

To bring structural information into the clustering without having to compute three-dimensional structures for 2,592 proteins, we used Foldseek's integration with ProstT5, a protein language model that predicts the 3Di alphabet directly from amino acid sequence. We built a Foldseek database from the combined VIR FASTA using ProstT5 to populate the 3Di field, then ran `foldseek easy-cluster` with mutual coverage 0.8, E-value 0.001, and `--alignment-type 2` (combined 3Di+AA scoring). This produced 73 clusters with three or more members – exactly half as many as the 146 sequence-based MCL clusters. Critically, the size distribution was now extremely skewed. The single largest cluster, which we labelled FS_001, contained 1,197 proteins, or 46% of the entire VIR repertoire across all three genomes. The next four clusters (FS_002 through FS_005) contained 385, 248, 143, and 136 proteins respectively. Beyond cluster ten, the clusters dropped quickly into the single digits.

We then asked, for each Foldseek cluster, how many sequence-based MCL subfamilies it merged. The answer was striking. FS_001 alone merged 89 of the 146 MCL subfamilies, FS_002 merged 36, FS_003 merged 32. In other words, the sequence-based pipeline was over-splitting the family by roughly a factor of two: many "subfamilies" that looked distinct by sequence were collapsed into single structural classes once the structural alphabet was taken into account. This is exactly the prediction Harrison and Higgins had made on the basis of the PIR crystal structure: that PIRs are structurally unified beyond the reach of sequence-based clustering.

When we counted protein membership rather than cluster identity, the picture of cross-continental conservation became even sharper. 97.3% of all 2,592 VIR proteins fall into structural clusters that contain members from all three reference genomes. The 47 Foldseek clusters that we initially flagged as "genome-unique" together account for only 52 proteins (2% of the total) – they are almost all singletons or pairs and likely represent rare lineage-specific variants on top of an otherwise universally conserved structural scaffold. Within the large universal clusters, the per-genome composition is roughly proportional to each genome's contribution to the dataset: in FS_001, for example, the 1,197 members break down as 400 from PvP01 (Indonesia), 349 from PvW1 (Thailand), and 448 from PvPAM (Peru), close to what one would expect if the underlying repertoires were drawn from a shared distribution.

0.4.4 4.4. Phylogeny at the family level (Plan C, Q1)

We then asked whether maximum-likelihood phylogenetic methods would tell the same story. Following the procedure used in the FoldMason paper for the Flaviviridae glycoproteins, we took the 73 Foldseek cluster representatives as input to FoldMason `structuremsa`, producing AA and 3Di multiple sequence alignments. We trimmed the alignments with trimAl. The procedure used by Gilchrist et al. was to trim columns with 35% or more gaps (`-gt 0.65`); we found that with our data this removed every column, because by construction the 73 cluster representatives are the most distant proteins in the family from each other. We instead used trimAl's `-automated1` mode, designed specifically for downstream ML phylogenetics on diverged data, which retained 749 of the original 1,521 columns and 70 of the 73 sequences. We then ran IQ-TREE 3 with ModelFinder for the AA alignment (which selected `VT+F+G4` as the best-fit model), with the Garg & Hochberg 2025 3Di substitution matrix (NEXUS format, obtained from the `jonasmehl1/structural_phylogenetics` repository, the same matrix used in the FoldMason paper) for the 3Di alignment, and with a partitioned analysis of concatenated AA+3Di. Each tree was inferred with 1000 ultrafast bootstrap replicates and 1000 SH-aLRT replicates for branch support.

The three trees agreed on the broad topology: a small number of well-supported deep clades each corresponding to a major group of structural classes, with the relationships between deep clades less well-supported. The trees are useful as a scaffold for downstream annotation – for example, asking which structural classes are most distant from the bulk of the family, or which subset of classes is restricted to particular life-cycle stages once expression data is layered on – but the small number of tips per branch and the high sequence divergence between cluster representatives means we should not expect strong resolution at every node.

0.4.5 4.5. Phylogeny within the dominant structural class (Plan C, Q2)

The more biologically interesting question is at finer resolution: within the dominant FS_001 fold, where 1,197 sequences from three continents coexist, is there any signal of geographic structure? To answer this we subsampled FS_001 with MMseqs2 at 50% sequence identity and 80% coverage, which yielded 338 sub-representatives (118 from PvP01, 110 from PvW1, 110 from PvPAM – excellent balance for an unbiased test). We aligned these with FoldMason, trimmed with `-gt 0.65` (which now worked: the cluster is homologous, so trimming retained 247 columns at 17% gap), and inferred three trees with IQ-TREE 3 using the same procedure as above. The AA model selected by ModelFinder was `VT+F+R6`.

We then tested whether genome of origin predicts position on the tree, by treating "genome of origin" as a three-state discrete character (PvP01, PvW1, PvPAM) and computing its parsimony score on each tree using phangorn. The minimum possible parsimony score, achieved if each genome formed a single monophyletic clade, is 2. A higher score means more state changes are needed, with perfect mixing (no geographic signal) giving the highest scores. To convert the score into a p-value, we randomized the tip labels 1000 times and recomputed the parsimony score on each shuffle.

The result was the same across all three trees. The observed parsimony score on the AA tree was 162, on the 3Di tree 156, and on the partitioned tree 160. The median parsimony score under random label shuffles was 160, with a range of roughly 145 to 177. The probability of observing a parsimony score at least as low as the actual one, by chance, was 0.69 for the AA tree, 0.26 for the 3Di tree, and 0.55 for the partitioned tree – none significant. Counting internal nodes whose descendants all came from a single genome gave 34 to 37 out of 336 internal nodes (10-11%), no more than expected by chance for three states distributed evenly across tips.

This is a clear negative result with a clear biological interpretation. Within the dominant VIR structural class, sequences from Indonesia, Thailand, and Peru are evolutionarily interleaved to the point of being indistinguishable from a random arrangement on the tree. The simple model in which each *P. vivax* lineage independently expanded and diversified its own VIR repertoire after the geographic separation of modern lineages is not consistent with these data. The data are more consistent with a scenario in which most of the diversification of FS_001 predated the geographic separation of *P. vivax* populations, and in which ongoing recombination and gene conversion among subtelomeric *vir* loci continue to keep the family panmictic across continents. The same general pattern – conservation of structural class assignment across geography, with sequence-level variation distributed randomly within those classes – holds for the other large structural clusters where we have enough sequences to test it.

0.4.6 4.6. What this exercise demonstrates

Three things are worth noting from a methods standpoint. First, the pipeline used by the FoldMason paper for the glycoprotein replication transfers cleanly to the *Plasmodium* VIR family with only one adjustment: the trimAl gap threshold used by Gilchrist et al. (`-gt 0.65`) was calibrated for closely related glycoprotein homologs and is too stringent when the input spans the entire diversity of a multigene superfamily; switching to trimAl's `-automated1` heuristic is the right choice in that case. Second, ProstT5 is good enough that no actual three-dimensional structure prediction is required: the entire structure-aware analysis can be done from FASTA input, which makes it tractable to apply to families of thousands of proteins on a laptop. Third, the combination of structural clustering (Foldseek + ProstT5) and structural phylogenetics (FoldMason + IQ-TREE with the Garg & Hochberg 3Di matrix) produces qualitatively different answers from the corresponding sequence-only pipelines, and in this case the structural answer is biologically more compelling: the VIR family is far more unified, and far less geographically structured, than its sequence diversity would suggest.